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Review article

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Cost-driven strategies towards industrially manufacturable perovskite photovoltaics

Ziyang Zhang (张子扬), Yingming Liu (刘英明), Zhenzhen Qin (秦臻臻), Mengjiong Chen (陈锰炯), Yanbo Wang (王言博) & Liyuan Han (韩礼元)

Abstract

With power conversion efficiencies exceeding 27%, perovskite solar cells (PSCs) are transitioning from laboratory-scale efficiency-driven research to a focus on large-scale deployment. At this stage, manufacturing cost is emerging as a key constraint governing the industrial competitiveness of PSCs. The manufacturing cost of PSCs (approximately US$0.3–0.6 W^{-1}) remains much higher than that of mature crystalline silicon solar cells (approximately $0.1 W^{-1}); therefore, it is essential to incorporate cost considerations into the future technological development of PSCs. In this Review, we discuss advances in PSC research from a cost-driven perspective, organized into three dimensions: enhancing manufacturing yield by addressing material reliability and process tolerance, reducing material costs by rethinking material selection and device architectures alongside material recycling strategies, and simplifying fabrication by alleviating dependence on controlled environments, specialized equipment and energy-intensive processing. Finally, we discuss how cost-oriented principles can be incorporated into PSC research and development while maintaining competitive device performance to facilitate their transition from laboratory studies to commercially viable photovoltaic technologies.

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State Key Laboratory of Metal Matrix Composites, Shanghai Jiao Tong University, Shanghai, China.
e-mail: han.liyuan@sjtu.edu.cn

Introduction

Photovoltaic (PV) technologies are being developed with the goal of delivering electricity in a long-term, stable and economically affordable manner to support the global energy transition[1]. The shift of these technologies from laboratory-scale performance validation to large-scale practical deployment represents an important real-world test. In this context, power conversion efficiency (PCE), operational stability and cost collectively define the 'golden triangle' for evaluating the practicality and competitiveness of PV technologies[2,3] (Fig. 1a).

Perovskite solar cells (PSCs) are a promising PV technology that combines high PCE with the potential for low-cost manufacturing[4,5]. At present, PSCs have achieved certified PCEs surpassing 27%[6,7] and operational lifetimes exceeding 10,000 h (ref. 8). These advances indicate that PSCs are approaching performance levels relevant to industrial application. Meanwhile, increasing industrial efforts in module-level validation and manufacturing scale-up suggest that PSCs are beginning to transition from laboratory research to commercialization. Although over ten 100-MW-scale production lines have been established, commercial deployment of PSCs remains limited, with existing shipments largely associated with demonstration projects[9]. This gap is largely attributed to the manufacturing cost of perovskite modules (approximately US$0.3–0.6 W^{-1})[10–12], which remains substantially higher than that of crystalline silicon modules (approximately $0.1 W^{-1})[13]. Therefore, it is necessary to consider how PSC costs are linked to materials, device architecture and manufacturing processes, so that cost considerations can be incorporated into technology development alongside efficiency and stability rather than being viewed solely as an economic metric.

Existing analyses consistently indicate that efficiency has an important role in reducing the manufacturing cost of PSCs. Higher efficiencies increase the energy output per unit area, reducing the required module area for a given power output and substantially lowering overall costs[14]. For instance, raising efficiency from 15% to 25% can reduce the manufacturing costs by more than a quarter[11,12,15–17]. Therefore, research efforts have focused on increasing efficiency using strategies such as crystallization regulation, interface engineering, defect passivation and the development of advanced charge transport layers (CTLs)[6,18–22]. However, even under assumptions of high module efficiency (that is, around 25%), the manufacturing cost of PSCs remains approximately two to three times higher than that of mature crystalline silicon technologies[12].

Additional factors must therefore be considered to address the cost challenges faced by PSCs, specifically manufacturing yield, material cost and fabrication processes[12,15,16,23,24] (Fig. 1b). Increasing manufacturing yield from 60% to 80% could reduce module cost by approximately 30% by increasing the effective use of materials[11,12,16,25] (Fig. 1c). Meanwhile, material costs account for the largest share of the manufacturing cost, typically approaching 70% (Fig. 1d), owing to the reliance on expensive materials for CTLs and electrodes[12,15,17]. In addition, PSC fabrication often requires strictly controlled environments, prolonged high-temperature annealing and vacuum-based deposition processes. Consequently, process-related costs account for over one quarter of the total manufacturing cost, with vacuum equipment and electricity consumption representing the largest contributors to capital and operating costs, respectively[12] (Fig. 1d). The close connection of these factors to material selection, device architecture and process design suggests that they should be explicitly considered in the development of PSC technologies while maintaining the competitiveness of device performance. Although numerous cost-reduction strategies have been proposed, they are often discussed separately across materials, device and manufacturing research.

In this Review, we examine the technological development of PSCs from a cost-oriented perspective, with particular emphasis on factors beyond efficiency. We first summarize the approaches for increasing manufacturing yield to minimize material loss by enhancing material reliability and process tolerance. Then, we review strategies to reduce material cost through low-cost material development, rational device architecture design, and material recycling and reuse. Next, we discuss efforts to simplify fabrication procedures by mitigating dependence on controlled environments, specialized equipment and energy-intensive processing. We aim to establish cost as a critical technical design parameter in material selection, device architecture and manufacturing processes and provide an integrated and application-oriented framework to support the translation of PSCs from laboratory demonstrations to industrially competitive technologies.

Increased manufacturing yield

Low manufacturing yield remains a bottleneck for the commercialization of PSCs, with reported module yields often falling short of industrially viable levels. Poor reproducibility is a common problem in large-scale manufacturing of PSCs, manifested by the accumulation of defects and pronounced batch-to-batch variations, which can increase the fraction of devices that fail to meet performance specifications, thereby reducing manufacturing yield and increasing material waste and cost[25]. Yield improvement is therefore an effective route towards cost reduction in PSC manufacturing. This section outlines strategies to increase manufacturing yield by addressing problems with raw-material purity, precursor-solution stability and process tolerance.

Material purity

The reproducibility of PSCs is highly sensitive to the purity of the materials used throughout the fabrication process. Even trace-level impurities can perturb crystallization pathways, introduce trap states and disrupt charge transport, amplifying batch-to-batch variations in film quality and device performance. Consequently, controlling impurity levels in perovskite precursors and charge transport materials (CTMs) can substantially increase manufacturing reproducibility.

Among commonly used perovskite precursors, PbI_2 represents an important source of impurity-related irreproducibility because commercial products synthesized via aqueous synthesis routes often contain residual water and impurity species[26,27]. Trace amounts of water can perturb precursor chemistry and film formation, leading to broad variations in film quality and device performance. For example, increasing water content in the PbI_2 precursor weakens the coordination between PbI_2 and dimethyl sulfoxide (DMSO), accelerating crystallization and promoting heterogeneous-dominated nucleation in the perovskite film during annealing[27] (Fig. 2a). Similarly, in organic iodide salts such as formamidinium iodide (FAI), trace metallic impurities such as Ca^{2+}, Na^{+} and K^{+} introduced during synthesis broaden device efficiency distributions[28]. In both cases, reducing precursor impurity levels improved reproducibility, with low-water-content PbI_2 yielding consistent performance across 11 devices[27] and reduced metallic impurity levels enabling consistent performance across 40 devices from eight batches[28].

The reproducibility of CTLs is also affected by impurity accumulation. Commercial C_{60}, a widely used CTM in inverted PSCs, can undergo gradual degradation during repeated thermal evaporation processes, resulting in the formation of higher-molecular-weight fullerene derivatives (such as C_{120} and C_{124}) that introduce interfacial trap states and cause batch-to-batch performance variability[29]. This issue can be mitigated by the sublimation purification of C_{60} before evaporation, which

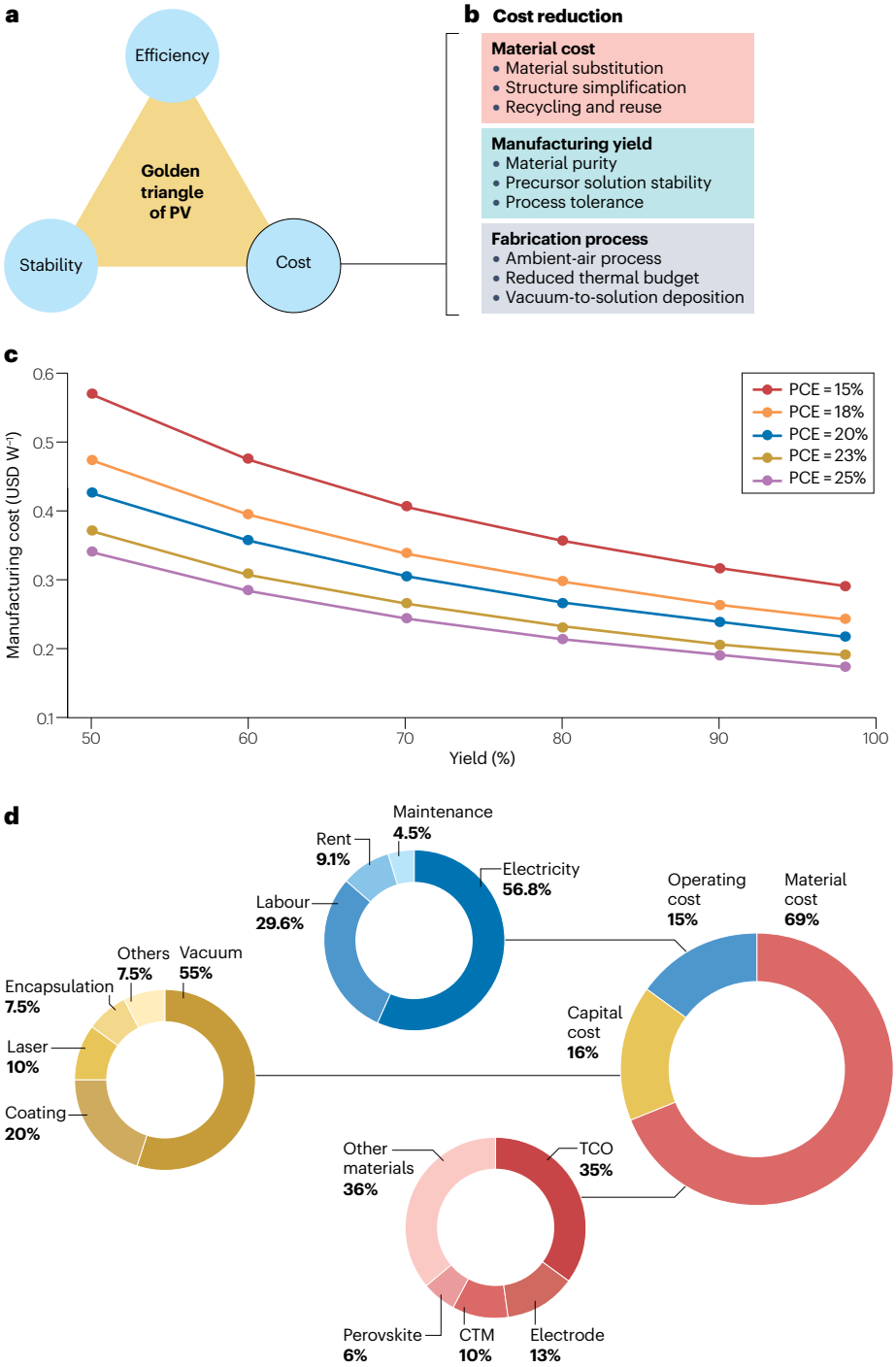
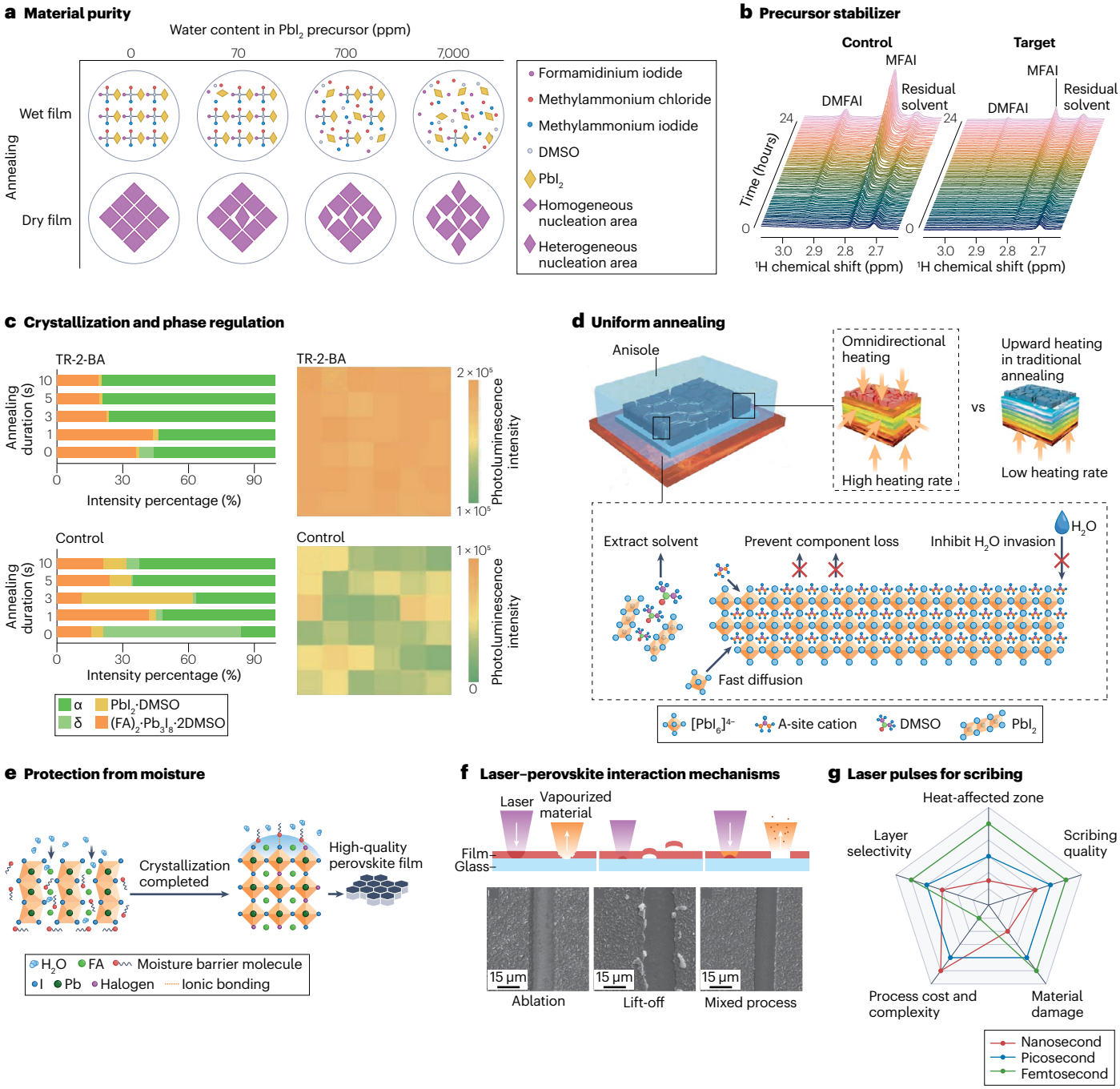


Fig. 1 | Factors influencing the commercialization of perovskite solar cells. a, The golden triangle of factors that influence the practicality and competitiveness of photovoltaic (PV) technologies. **b**, Three complementary pathways for reducing the cost of perovskite solar cells (PSCs): manufacturing yield improvement, material cost reduction and fabrication process simplification. **c**, Sensitivity analysis showing the influence of manufacturing yield on the manufacturing cost of PSCs at different power conversion efficiencies (PCEs). **d**, Representative overall cost composition of PSC manufacturing and breakdown of the contributions to the total material, capital and operating costs (data from ref. 12). CTM, charge transport material; TCO, transparent conducting oxide. Panel **c** adapted from ref. 12, CC BY 4.0.

removes residual oxygen and suppresses subsequent coalescence during repeated deposition[29].

Stability of perovskite precursor solutions

The intrinsic chemical instability of perovskite precursor solutions (PPSs), which typically comprise Pb halides, cation salts and coordinating solvents, is another barrier to reproducible device fabrication. During storage, mixed-cation solutions undergo iodide oxidation to I_2 or I_3^-, and organic cations deprotonate and react to form large by-products such as *N*-methylformamidinium iodide (MFAI) and *N,N*-dimethylformamidinium iodide (DMFAI)[30,31]. These concurrent processes shorten the shelf life of the PPS, disrupt perovskite crystallization and film formation and ultimately cause pronounced batch-to-batch variability in device performance.



A widely adopted approach to stabilize PPSs is the incorporation of functional additives that suppress chemical degradation during storage. Additives capable of regulating the redox chemistry of iodine species, including hydrazine-based compounds, such as benzylhydrazine hydrochloride (BHC), and phosphonate-based compounds, such as diethyl(hydroxymethyl) phosphonate (DHP), inhibit iodide oxidation and mitigate the accumulation of oxidized iodine species[32,33]. By suppressing redox-induced degradation, these additives effectively prolong the shelf life of the PPS to around 2 months and reduce the trap formation associated with solution ageing.

In parallel, additives that stabilize organic cations through hydrogen bonding and local acid–base regulation can suppress cation deprotonation and the formation of irreversible by-products, further increasing PPS stability. Additives such as piperazine-2-carboxylic acid dihydrochloride (PCADCl), 2,3,5-trichlorobenzaldehyde (3Cl-BZH), Lewis-basic ionic-liquid additives and hydrazine-based acids can stabilize methylammonium (MA^+) or formamidinium (FA^+) cations while maintaining a mildly acidic microenvironment[30,34–36] (Fig. 2b). For example, the Lewis-basic ionic-liquid 1,3-bis(cyanomethyl)imidazolium chloride ([Bcmim]Cl) suppresses MA^+ deprotonation, shifting the onset

Fig. 2 | Improving manufacturing yield of perovskite solar cells. a, Increasing water content in a PbI_2 precursor reduces the coordination bonding between PbI_2 and dimethyl sulfoxide (DMSO), leading to heterogeneous nucleation in the dry film after annealing. **b**, ^{1}H NMR spectra (normalized to the residual solvent peak) of a perovskite precursor solution with (right) and without (left) a Lewis-basic ionic-liquid additive, plotted at 15-min time intervals. The additive reduces the formation of decomposition by-products (*N*-methylformamidinium iodide (MFAI) and *N*,*N*-dimethylformamidinium iodide (DMFAI)), indicating increased precursor stability. **c**, Left, intensity percentage of perovskite and intermediate phases from X-ray diffraction (XRD) measurements with increasing annealing duration, showing that tris(2-benzimidazolylmethyl)amine (TR-2-BA) suppresses the formation of PbI_2•DMSO and the non-photoactive δ-phase, enabling a simplified phase-evolution pathway via the $(FA)_2Pb_3I_8$·2DMSO intermediate phase towards the photoactive α-phase. Right, photoluminescence intensity maps of 36 films demonstrate that TR-2-BA also increases batch-to-batch reproducibility. **d**, Liquid medium annealing achieves high-rate omnidirectional heating by using preheated anisole in direct contact with the perovskite film, in contrast to traditional annealing in which heat is transferred upwards through the substrate. Anisole extracts residual DMSO and the rapid heating promotes reactant diffusion, and the liquid medium suppresses volatile-component loss and moisture ingress. **e**, A crystallization-activated moisture barrier, produced by adding dodecyltrimethylammonium bromide (DTAB) to the perovskite precursor. During initial annealing, moisture enters the film and promotes crystallization; DTAB then migrates to the surface to form a dense hydrophobic barrier against further moisture ingress. **f**, Top, schematic illustration of the removal of perovskite film material through local melting and vaporization of the film caused by laser ablation, lifting off the film from the substrate using vapour pressure generated by interfacial heating and a combination of the two processes. Bottom, scanning electron microscope images of a $CH_3NH_3PbI_3$ ($MAPbI_3$) perovskite film following material removal using each of these three approaches, with the mixed process producing the cleanest scribe. **g**, Radar chart comparing nanosecond, picosecond and femtosecond laser pulses for scribing perovskite solar modules[74]. A larger radius indicates greater favourability, including lower cost and complexity, less material damage and smaller heat-affected zone. FA, formamidinium. Panel **a** reproduced from ref. 27, CC BY 4.0. Panel **b** reproduced from ref. 36, CC BY 4.0. Panel **c** reproduced with permission from ref. 39, Wiley. Panel **d** reproduced with permission from ref. 60, AAAS. Panel **e** reproduced with permission from ref. 64, AAAS. Panel **f** adapted from ref. 72, Springer Nature Limited.

of MFAI formation from 30 °C to 55 °C and allowing the PPS to continue to form high-quality films after 10 days of ageing[36]. The effectiveness of hydrazine-based molecules in stabilizing PPSs depends on the type and relative position of weakly acidic functional groups, which can regulate molecular acidity. This relationship provides useful molecular design principles[37]. Through these approaches, the shelf life of PPSs can be extended from only a few days to several weeks or even months, while maintaining narrow device performance distributions and improving batch-to-batch reproducibility.

The solvent system also influences PPS stability by promoting or suppressing solvent-dependent side reactions in mixed-cation PPSs containing FA^+ and MA^+ (ref. 38). Protic solvents such as 2-methoxyethanol (2-ME) and isopropanol (IPA) facilitate MA^+ deprotonation and promote the formation of MFAI and NH_4Cl, whereas water exhibits more complex behaviour, promoting MFAI formation at high temperatures while inducing its hydrolysis into formamide and formic acid[38]. Among aprotic solvents, *N*,*N*-dimethylformamide (DMF) shows higher chemical reactivity owing to the formation of dimethylamine (DMA^0) upon hydrolysis[38], which catalyses side reactions in mixed-cation solutions, whereas DMSO and γ-butyrolactone (GBL) are comparatively less reactive towards such side reactions[38]. Collectively, these solvent-dependent effects contribute to batch-to-batch variability across different PPS formulations and highlight the importance of solvent selection for reproducible PSC fabrication.

Process tolerance

In addition to material purity and PPS stability, the fabrication process is a major source of yield variability. The fabrication of PSCs involves complex solution-based film formation processes, strong interactions with external environments and narrow process windows. Consequently, even small deviations during solution-based formation processes, including perovskite crystallization and molecular interfacial assembly, or during interconnection and post-treatment can propagate and accumulate across multiple fabrication steps, resulting in pronounced variations in device performance. The focus of PSC fabrication has therefore expanded beyond identifying optimal conditions for individual steps towards broadening overall process tolerance to enable reproducible manufacturing.

Solution-based film formation. During perovskite crystallization and film formation, multiple Pb–solvent complexes and intermediate phases often coexist, and rapid solvent evaporation and non-uniform mass transfer introduce strong kinetic perturbations, leading to uncontrolled crystallization and poor reproducibility.

Regulating the evolution of intermediate phases can increase the robustness of crystallization processes. By stabilizing specific precursor–solvent complexes and suppressing the formation of competing intermediates through precursor additives such as tris(2-benzimidazolylmethyl)amine (TR-2-BA), crystallization pathways can be directed towards more deterministic phase evolution (Fig. 2c), reducing sensitivity to processing conditions and improving reproducibility[39]. Uncontrolled nucleation can also be mitigated by moderating solvent-removal kinetics. Introducing a 2-min ageing step at room temperature under N_2 before annealing, which provides sufficient time for the gradual removal of residual solvents, allows partial pre-crystallization and more gradual phase transformation, narrowing performance distributions across batches[40].

Increasing mass transfer uniformity is also important to enable the formation of uniform large-area films. Non-uniform solvent evaporation can induce local morphological instabilities such as coffee-ring effects and void formation, which amplify spatial variations in perovskite nucleation and growth[41]. Strategies such as in situ polymer frameworks or semi-sealed gas quenching[41,42] suppress lateral solute migration and enable uniform crystallization over large areas, leading to reduced area-dependent performance losses and improved batch-to-batch reproducibility during scale-up.

Similar processing challenges also exist for self-assembled monolayer (SAM) hole-selective layers fabricated by solution processing, which are increasingly used in high-efficiency inverted PSCs[43]. SAM formation relies on molecular adsorption and assembly during solution deposition; therefore, variations in solvent properties, solution concentration and substrate surface conditions that disturb molecular anchoring and packing can lead to incomplete coverage, local aggregation or interfacial disorder. These effects are further amplified during the fabrication of large-area SAMs because maintaining uniform SAM coverage and ordered molecular assembly across extended substrates is even more challenging.

Various strategies have been developed to improve the controllability and reproducibility of SAM assembly. Solvent polarity can strongly influence the aggregation configuration of SAM molecules, as the coordinating interaction between solvent and substrate determines the ability of SAM molecules to effectively approach and anchor onto the substrate surface[44]. Therefore, solvent systems and solution concentration can be regulated to suppress molecular aggregation during SAM formation[45]. Another common strategy is substrate engineering, including surface activation, pretreatment, and the use of metal-oxide substrates such as NiO_x that provide favourable anchoring sites and increase the uniformity of molecular assembly at the substrate–SAM interface[46–50]. Co-assembly and mixed-SAM strategies have also attracted increasing attention, in which auxiliary molecular components are introduced to fill uncovered regions, regulate intermolecular interactions and improve molecular dispersion, mitigating local aggregation and interfacial disorder[22,51–55]. Furthermore, molecular-level design of SAM materials, including optimization of anchoring groups, tuning of molecular polarity, and enhancement of intermolecular interactions and molecular packing, can also improve process tolerance and large-area manufacturability of SAM hole-selective layers[56–59].

Environmental sensitivity. Beyond the intrinsic challenges associated with solution-based film formation, the reproducibility of PSC fabrication is strongly constrained by fluctuations in external environments, particularly temperature and humidity, which often cause inconsistent results across batches, laboratories and even seasons.

Temperature fluctuations primarily influence crystallization by modifying solvent evaporation rates and the stability of intermediates during annealing. To mitigate this sensitivity, strategies that buffer thermal gradients and provide well-defined thermal environments have been developed. For example, the liquid medium annealing strategy uses a preheated liquid medium to provide omnidirectional heating, minimizing temperature gradients and suppressing the loss of volatile components[60] (Fig. 2d). By decoupling crystallization behaviour from ambient temperature variations, this strategy promotes more consistent phase conversion and narrows performance distributions across different seasonal conditions.

Humidity is another crucial environmental variable. Moisture can participate in multiple stages of film formation, influencing nucleation density, ion mobility and defect healing, but excessive or poorly regulated humidity promotes parasitic phase formation and compositional inhomogeneity[61,62]. These challenges can be mitigated either by operating within well-defined humidity windows during processing[61,62] or by adopting humidity-tolerant fabrication methods that actively regulate moisture participation during crystallization[63]. For example, allowing moisture ingress during initial annealing promotes the formation of the photoactive α-phase, and a subsequently formed moisture barrier suppresses excess moisture after crystallization[64] (Fig. 2e). This approach extends the allowable relative humidity range for PSC fabrication to 20–93%, enabling more robust and reproducible fabrication under practical environmental conditions.

Narrow process windows. Even under well-controlled film quality and processing environments, many fabrication steps in PSC manufacturing remain intrinsically sensitive to process parameters. Consequently, small deviations from optimal conditions can lead to pronounced variability in film quality and device performance.

Conventional perovskite surface passivation requires that the concentration of the passivator is within a tightly defined range. The desired concentration range can vary substantially across different perovskite compositions and device architectures. Insufficient passivator concentration leaves interfacial defects unhealed, whereas excessive passivation induces undesired low-dimensional phases and hinders charge transport[65]. This sensitivity can be alleviated using saturated passivation strategies, in which excess passivators are removed in a post-treatment step, enabling effective passivation across a wide concentration range and different material systems[65].

Quenching during perovskite film formation also has a narrow tolerance. Conventional quenching methods are highly sensitive to parameters such as pumping speed, solvent composition and treatment duration, resulting in unstable nucleation dynamics[66]. Quenching strategies with broadened parameter tolerance have been developed to reduce sensitivity to precise operational conditions and to suppress batch-to-batch performance variations. For example, temperature-controlled vacuum quenching uses substrate cooling to prolong intermediate-phase lifetimes and stabilize nucleation[66].

Process tolerance challenges also arise in the vacuum-based deposition of perovskites because small variations in particle size or vapour pressure can lead to large composition deviations. One route to address this issue is the molten salt sequential evaporation strategy, in which mixed halide precursors form a liquid phase with uniform thermal transport and stable evaporation rates, enabling reproducible composition control[67]. A complementary evaporation–solution hybrid route deposits a uniform PbI_2-based scaffold by evaporation, followed by solution-derived organic halides, combining the thickness uniformity of evaporation with more complete conversion during solution processing to suppress batch-to-batch variation and reduce scale-dependent performance losses[68].

Laser scribing-induced damage. The fabrication of perovskite solar modules relies on the series interconnection of sub-cells, which is typically implemented through three sequential laser scribing steps[69,70]. Owing to the intrinsically soft and defect-prone nature of perovskite films, localized laser irradiation can easily induce perovskite decomposition, interfacial delamination and shunt formation, compromising module efficiency, stability and reproducibility[4,71–73].

Laser–perovskite interactions can proceed through different material removal pathways, including direct ablation, in which irradiated material is vapourized; interfacial lift-off, in which the overlying film is detached from the interface; and mixed mechanisms combining the two processes[72] (Fig. 2f). The pathway through which material is removed depends on where the laser energy is absorbed and how quickly heat spreads within the device stack, which are primarily determined by laser pulse duration, wavelength and fluence[71]. Nanosecond lasers have relatively long interaction times and therefore allow substantial thermal diffusion and typically generate wide heat-affected zones, often accompanied by perovskite decomposition and rough scribe edges. Picosecond lasers have shorter energy deposition times, reducing thermal damage and producing cleaner trenches, and femtosecond lasers enable nearly 'cold' ablation through ultrafast, non-thermal processes, yielding minimal collateral damage. However, the industrial application of femtosecond lasers is limited by their high cost and low throughput, rendering nanosecond and picosecond lasers more practical for large-area module fabrication[74] (Fig. 2g). Beyond pulse duration, wavelength selection determines which layer predominantly absorbs the laser energy[72]. Additionally, laser fluence and scanning conditions (such as scanning speed and pulse overlap) need to balance incomplete material removal against excessive thermal

damage. Together, these factors render laser scribing highly sensitive to small variations in processing parameters[73,75,76].

To mitigate laser scribing-induced damage, various strategies have been developed to limit heat generation in the perovskite layer and confine damage to well-defined regions rather than relying solely on precise parameter control. For example, the back-side irradiation strategy initiates energy deposition at buried interfaces rather than within the perovskite absorber[74,77–79]. In addition, plume-management or debris-management strategies, such as gas-flow-assisted[80] or liquid-assisted scribing[81], remove ablation by-products and suppress thermal accumulation. By reducing debris formation and perovskite decomposition, these methods yield cleaner scribe lines and higher process reproducibility. Beyond laser-based optimization, alternative patterning concepts have also been explored to avoid laser-induced damage altogether. For instance, mask-assisted deposition strategies can directly define interconnection regions without ablation, eliminating localized heating, debris formation and shunt pathways associated with laser scribing[82,83]. Collectively, these approaches aim to alleviate the yield limitations of laser-based interconnection and improve process tolerance and robustness.

Reduced material cost

The high cost of materials represents another major barrier to cost-effective PSCs[12,15,23]. Although the estimated contribution of material costs varies across analyses owing to differences in underlying assumptions and manufacturing routes, material costs generally account for a major share of overall manufacturing cost. Transparent conducting oxides (TCOs), CTMs and noble-metal electrodes together account for over half of the total material cost of PSCs, primarily owing to their reliance on scarce raw materials and the need for complex synthesis or deposition processes[12,15,17,84,85] (Fig. 1d). By contrast, the perovskite absorber layer itself contributes only a minor fraction (approximately 6%). Although other module-level materials, including encapsulation materials, aluminium frames and junction-box-related materials, together contribute approximately 36% to the overall material cost, their manufacturing processes and supply chains are largely shared with mature PV technologies, resulting in comparatively low cost uncertainty[16]. Accordingly, efforts to reduce the material costs of PSCs should focus on materials that not only make a large contribution to overall module cost but also have potential for further PSC-specific cost reduction. Therefore, research is primarily focused on PSC-specific material cost contributors, including TCOs, CTMs and noble-metal electrodes, and recycling strategies to prolong the lifetime of these high-value components. This section outlines material substitution, structure simplification and recycling and reuse strategies to reduce material costs.

Material substitution

The typical costly functional materials used in common PSCs include organic hole transport materials (HTMs), such as 2,2′,7,7′-tetrakis(*N*,*N*-di-*p*-methoxyphenylamine)-9,9′-spirobifluorene (spiro-OMeTAD); fullerene-based electron transport materials (ETMs), for example, C_{60} and [6,6]-phenyl-C_{61}-butyric acid methyl ester (PCBM); noble-metal electrodes, such as Au and Ag; and TCOs, for example, fluorine-doped tin oxide (FTO) and indium tin oxide (ITO). Substituting these costly components with abundant and inexpensive alternatives has been widely explored as a direct strategy to reduce the material cost of PSCs. However, as these functional layers are crucial to charge transport and interfacial stability in PSCs, material substitution is often accompanied by performance and durability challenges.

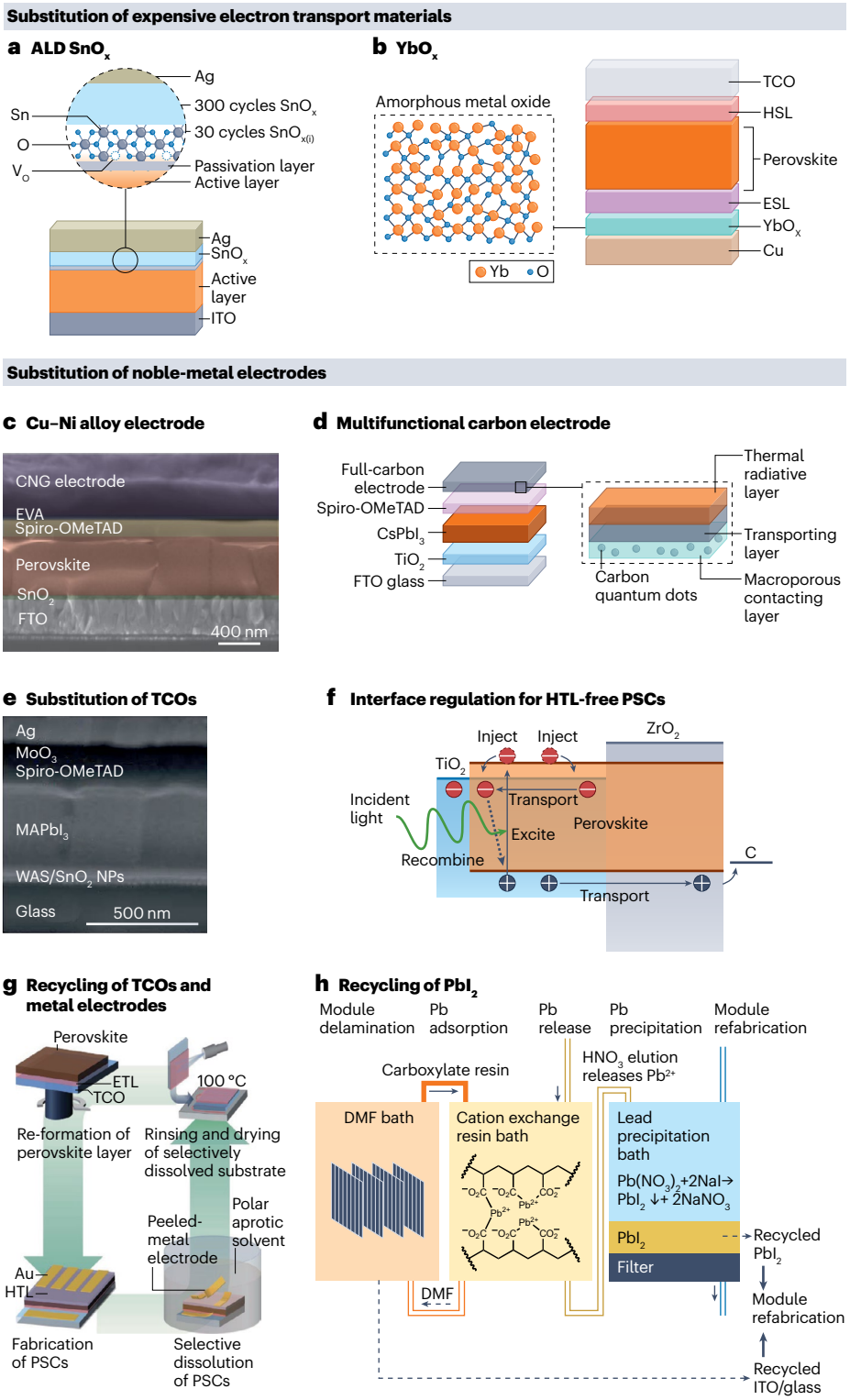
CTMs. For HTMs, the widely used spiro-OMeTAD can be replaced by inorganic semiconductors such as CuSCN[86,87], CuI[88] and $CuGaO_2$ (ref. 89) or by low-cost organic HTMs, including organic HTMs based on simplified molecular backbones[90,91]. Additionally, spiro-core derivatives that are synthesized in a few steps from readily available precursors can also be used[92]. However, further optimization and validation are still needed to achieve performance fully comparable to spiro-OMeTAD across different PSC systems.

SnO_x is one of the most promising candidates for non-fullerene-based electron transport layers (ETLs) owing to its energy-level alignment with perovskite, high electron mobility and high chemical stability. Additionally, the cost of SnO_x is over 50% less than that of fullerene derivatives, making it an attractive candidate for low-cost ETLs[93,94]. To achieve highly uniform and conformal SnO_x layers on the thermally sensitive perovskite surface, deposition techniques such as atomic layer deposition (ALD)[48] (Fig. 3a) or hybrid solution–ALD routes[95] are commonly used to accommodate the limited thermal budget of the perovskite layer. However, these techniques can lead to increased process complexity. One example of a non-ALD route for fabricating functional metal-oxide layers is the formation of ytterbium oxide (YbO_x) by spontaneous oxidation of thermally evaporated Yb, which serves as a buffer layer between the ETL and metal electrode (Fig. 3b), while maintaining efficient electron extraction and interfacial protection[96].

Noble-metal electrode. Owing to its high earth abundance and low cost, Cu was among the earliest earth-abundant candidates explored to replace conventional noble-metal electrodes[97]. However, the stability and operational lifetime of Cu electrodes can be compromised by halide-induced corrosion and metal ion diffusion. Interfacial protection strategies have been developed to mitigate these issues, including corrosion-inhibiting layers based on benzotriazole complexes[98] and composite electrode designs using Cu–Ni alloys encapsulated by graphene barriers to suppress ion diffusion and moisture ingress[99] (Fig. 3c). These strategies have enabled Cu-based electrodes to achieve long-term operational stability as a step towards their practical application.

Beyond Cu, other low-cost metals and alloys provide additional routes to reduce electrode cost. Molybdenum-based composite electrodes leverage intrinsic chemical inertness and strong resistance to halide diffusion, achieving stable device operation when combined with adhesion-promoting interlayers[100]. Additionally, low-melting-point alloys, such as In–Sn–Bi alloys, can be used for stable blade-printable electrodes that solidify at room temperature without vacuum processing while offering recyclability through remelting and reuse[101].

Carbon-based electrodes are another important class of noble-metal alternatives with low cost and high abundance and chemical stability. Early carbon-based devices were limited by poor interfacial contact and unfavourable energy-level alignment, often resulting in open-circuit voltage (V_{OC}) loss, reduced fill factor and low overall device efficiency[102]. To address these issues, interface regulation strategies have been developed, including solvent-assisted crystallization[103], self-adhesive macroporous carbon architectures[104], work-function tuning via a Ti_1–graphene single-atom material[105] and dipole-coupled carbon nanotube interfaces[106]. Additional approaches that introduce efficient charge-selective interlayers, such as hole transport bilayers[107], Cu(II) phthalocyanine hole transport layers (HTLs)[108] and gradient ETL architectures[109], further suppress associated interfacial recombination losses. Multifunctional three-layer carbon electrodes have also been designed, comprising a macroporous carbon layer modified with carbon quantum dots to improve interfacial contact, a conductive graphite



layer to facilitate charge transport and a dense carbon layer to promote radiative cooling[110] (Fig. 3d). Together, these functions improve device performance and operational stability, highlighting the potential of multilayer carbon electrodes for durable and low-cost PSCs[110].

TCOs. ITO and FTO remain the most widely used transparent electrodes in PSCs, but their reliance on scarce elements or high-temperature processing results in high material costs, motivating the development of low-cost alternatives.

Fig. 3 | Reducing the material cost of perovskite solar cells. a, A perovskite solar cell (PSC) structure using SnO_x prepared by atomic layer deposition (ALD) as a low-cost electron transport layer (ETL), with a 30-cycle SnO_x(i) interlayer for interfacial oxygen-vacancy (V_O) regulation and a 300-cycle SnO_x layer for electron transport. **b**, An inverted PSC structure with a YbO_x buffer layer between the Cu electrode and electron-selective layer (ESL) formed by spontaneous oxidation of thermally evaporated Yb, which facilitates electron transport and suppresses the interdiffusion of harmful species. **c**, Cross-sectional scanning electron microscopy image of a PSC with a rear electrode comprising a Cu–Ni alloy encapsulated with graphene on both sides (CNG electrode) to increase the stability of the electrode. **d**, A PSC structure that uses a triple-layer carbon electrode, comprising a dense carbon layer, which dissipates heat, a graphite charge transporting layer and a macroporous contacting layer, which provides close contact with the hole transport layer (HTL) and is modified with carbon quantum dots to improve interfacial energy alignment. **e**, Scanning electron microscope image of a PSC that uses an oxide–metal–oxide architecture – WO_3/Ag/SnO_2 (WAS) – as a transparent conducting oxide (TCO) alternative. The WAS layer is modified with SnO_2 nanoparticles (NPs) to facilitate band-level matching between the WAS and the $CH_3NH_3PbI_3$ ($MAPbI_3$) perovskite layer. **f**, Schematic illustration of energy-level alignment in a printed HTL-free PSC with a triple-layer mesoscopic architecture. Photoexcited electrons are selectively injected into TiO_2, whereas holes are transported through the perovskite to the carbon electrode; the intervening ZrO_2 layer suppresses back-contact recombination. **g**, PSC recycling using a polar aprotic solvent to dissolve the perovskite and HTL, allowing the metal electrode to peel away for recovery and leaving the ETL-coated substrate for reuse in a new PSC. **h**, A process to recycle perovskite solar modules by delaminating the modules so the indium tin oxide (ITO)/glass can be reused and dissolving the perovskite in *N*,*N*-dimethylformamide (DMF), followed by Pb^{2+} adsorption by a cation-exchange resin, release into aqueous solution as $Pb(NO_3)_2$ using HNO_3 and precipitation as PbI_2 with NaI for module refabrication. EVA, ethylene vinyl acetate; FTO, fluorine-doped tin oxide; HSL, hole-selective layer; spiro-OMeTAD, 2,2′,7,7′-tetrakis(*N*,*N*-di-p-methoxyphenylamine)-9,9′-spirobifluorene. Panel **a** reproduced with permission from ref. 48, AAAS. Panel **b** reproduced from ref. 96, Springer Nature Limited. Panel **c** reproduced from ref. 99, Springer Nature Limited. Panel **d** reproduced from ref. 110, CC BY 4.0. Panel **e** reproduced with permission from ref. 114, RSC. Panel **f** reproduced with permission from ref. 122, AAAS. Panel **g** reproduced from ref. 139, CC BY 4.0. Panel **h** reproduced from ref. 141, CC BY 4.0.

Carbon-based transparent electrodes, including graphene and carbon nanotubes, have been explored as possible TCO alternatives owing to their potential for lower costs while maintaining high optical transparency and good chemical stability. Although conductivity enhancement and interface engineering have improved device performance, the efficiencies of carbon-based transparent electrodes remain lower than those of their ITO-based counterparts[111,112]. Therefore, further improvements in the charge transport and interfacial characteristics of carbon-based transparent electrodes are needed.

Alternatively, oxide–metal–oxide architectures combine the high conductivity of ultrathin metal layers with the optical transparency and charge selectivity of oxide layers, enabling ITO-free PSCs through rational compositional design and interfacial engineering[113,114]. For example, in WO_3/Ag/SnO_2 electrodes (Fig. 3e), an additional SnO_2 nanoparticle interlayer was required to improve energy-level matching and interfacial electron extraction[114]. Additionally, indium-free oxide electrodes, such as aluminium-doped zinc oxide (AZO), combine optical transparency with electrical conductivity and are estimated to have material costs over 50% lower than those of ITO[93]. By replacing the high-temperature processing conventionally required to achieve sufficient conductivity with room-temperature pulsed laser deposition, AZO electrodes can be fabricated without substrate heating or post-deposition annealing[115]. Nevertheless, compared with high-quality ITO (>90% transmittance and approximately 10^6 S m^{-1} conductivity), AZO (>80% transmittance and 1.8×10^5 S m^{-1} conductivity) still requires further improvements in conductivity and optical transparency to become a more competitive alternative[115].

Structure simplification

In PSC architectures, the perovskite absorber layer is typically sandwiched between an ETL and an HTL to facilitate charge extraction and interfacial regulation. Although these CTLs are important for achieving high device performance, they often contribute substantially to the overall material cost of PSCs. Given the ambipolar transport nature and long carrier diffusion lengths of perovskites, simplified architectures such as ETL-free and HTL-free PSCs have been explored as possible pathways towards reducing material costs[116]. However, removing the ETL or HTL inevitably introduces challenges, including poor perovskite crystallization, severe interfacial recombination, energy-level mismatch, unfavourable electrode contacts and insufficient stability.

For CTL-free PSCs, one central challenge arises during perovskite crystallization, as direct growth on TCO substrates often leads to poor wettability and consequently to non-uniform nucleation and buried interfacial defects. Therefore, surface pretreatment and crystallization-regulation strategies have been widely adopted[117–120]. For example, potassium fluoride (KF) treatment improved the wettability of an FTO surface and promoted heterogeneous nucleation, enabling the formation of dense and uniform perovskite films with reduced interfacial defect density[117]. In addition to strategies to control initial nucleation, post-crystallization approaches such as moisture-assisted secondary recrystallization[118], grain-boundary reconstruction via electrophilic reaction[119] and chemically activated interface reconstruction[120] further enlarge grain sizes and heal interfacial voids.

Beyond film formation, the absence of selective transport layers also leads to insufficient charge selectivity at the perovskite–electrode interface, which commonly results in substantial V_{OC} losses. Various interfacial engineering strategies have therefore been developed to restore charge selectivity and enhance charge extraction. These strategies include the insertion of ultrathin tunnelling or buffer layers that selectively transport charges[121,122] and the introduction of internal electric fields through compositional gradients[123–125], surface modification[126–128] and low-dimensional interfacial layers[129–131]. For example, an HTL-free mesoscopic TiO_2/ZrO_2/carbon architecture enables selective electron injection into TiO_2 and hole transport to the carbon electrode, while the intervening ZrO_2 layer suppresses back-contact recombination[122] (Fig. 3f). Collectively, these approaches have made CTL-free architectures increasingly viable, although they have not yet fully replaced the performance benefits provided by conventional CTLs.

Stability is another concern in CTL-free PSCs because direct contact between the perovskite and electrodes exacerbates ion migration, interfacial reactions and environment-induced degradation. Therefore, chemically inert and hydrophobic electrodes, particularly carbon-based contacts, have been widely adopted to suppress interfacial corrosion and enhance long-term operational stability[102,132–134]. Additionally, chemical passivation strategies have been used to inhibit ion migration and stabilize interfaces in CTL-free PSCs[128,135].

Recycling and reuse

In addition to reducing material costs through low-cost material substitution or simplified device architectures, recycling and reuse represent important complementary strategies. During manufacturing, PSC production inevitably generates defective devices and process scrap, and at the end-of-life stage, retired modules constitute an additional source of recoverable high-value components. Key constituents such as noble-metal electrodes and TCO substrates possess economic value, and Pb-based perovskites often raise environmental concerns, highlighting the relevance of their recovery and reuse for cost reduction and sustainability[136,137]. Integrated recycling and remanufacturing strategies that combine recovered PSC components with fresh feedstocks could enable the preservation of over 70% of the material value in PSC modules[136].

The selective dissolution of functional layers enables the reuse of TCO substrates and electrodes. In this approach, organic CTLs and perovskite absorbers are removed using solvents that dissolve the respective layers, facilitating the detachment of noble-metal electrodes with low residual Pb contamination and preservation of the underlying TCO surface[138–140] (Fig. 3g). The recycled TCO substrates can maintain their photovoltaic performance over up to ten reuse cycles[139].

The recovery and reuse of Pb from perovskite absorbers address issues of economic value and environmental safety. Pb recovery strategies typically combine solvent-assisted dissolution with ion-exchange-based sequestration (Fig. 3h), enabling overall Pb recovery ratios of over 99% and recovery of high-purity PbI_2 for device refabrication[141]. Devices fabricated from PbI_2 recycled through this process exhibit photovoltaic performance comparable to those based on fresh commercial PbI_2 (ref. 141). Alternatively, aqueous-based, layer-by-layer routes allow dissolution of perovskite materials followed by recrystallization and the sequential recovery of multiple device components, including CTMs, transparent electrodes and metal electrodes. Such approaches enable holistic multi-round recycling with recovery ratios exceeding 96% for several recovered components and refabricated devices that retain over 99% of their initial efficiency after five degradation–recycling cycles[142].

Beyond component-level recycling, module-level reuse and remanufacturing strategies have also been explored to mitigate the shorter operational lifetimes of PSCs relative to crystalline silicon solar cells. Periodic rejuvenation concepts involve the removal of the degraded device stack above the ITO substrate and subsequent deposition of a fresh device stack onto the cleaned ITO substrate[143]. The performance of these refabricated devices does not substantially degrade over multiple rejuvenation cycles, demonstrating the feasibility of this approach[143].

Collectively, these recycling and reuse strategies highlight viable routes towards cost-effective and sustainable perovskite PVs. Nevertheless, achieving sufficiently pure recovered materials might involve additional purification steps that introduce further processing complexity or cost. Additionally, the need to manage and recover solvents used during recycling and to address compositional diffusion across functional layers in end-of-life devices might further complicate material recovery[137]. Therefore, further developments are needed to enable the large-scale implementation of these strategies.

Simplified fabrication process

Fabrication simplification is another pathway towards lowering the manufacturing cost of PSCs. Conventional fabrication routes typically rely on controlled N_2 gloveboxes to exclude moisture and oxygen, thermal annealing to control crystallization kinetics, and vacuum-based deposition to enable conformal deposition of CTLs, all of which substantially increase capital costs and energy consumption. Simplified fabrication routes can directly reduce dependence on specialized environments and equipment and reduce energy demand, contributing to overall cost reduction[16,17,144], provided that device performance, stability and process reproducibility can be maintained. This section outlines approaches to reduce process costs using ambient-air processes, reduced thermal budget and vacuum-to-solution deposition.

Ambient-air process

Ambient-air fabrication is less reliant on specialized facilities and inert environments than glovebox-based processing. However, exposure to oxygen and moisture can induce defect formation and undesirable phase transitions in perovskite films. Additionally, rapid solvent evaporation and the hygroscopic nature of commonly used solvents often induce inhomogeneous morphology in the perovskite film. Conventional solvent systems are also often toxic and environmentally hazardous, raising additional safety and environmental concerns for ambient-air fabrication[145]. These challenges have long hindered the feasibility of air processing. Progress in material design, solvent engineering and interface regulation has allowed devices produced with ambient-air processing to achieve efficiencies approaching those of devices fabricated in controlled environments, suggesting that the feasibility of ambient-air processing for low-cost PSC manufacturing is growing.

Oxygen and moisture. During ambient-air processing, oxygen and moisture can interfere with perovskite crystallization and stability through hydration, oxidation and deprotonation reactions, leading to defect formation, phase transitions and accelerated degradation[146,147]. Defect passivation has emerged as an effective and widely adopted strategy to suppress redox reactions and inhibit moisture-induced decomposition. Passivator additives coordinate Pb^{2+} ions and stabilize organic cations, reducing the density of defect sites susceptible to reactions with oxygen and moisture and promoting more controlled crystallization in ambient air[18,31,148–151]. For example, guanabenz acetate (GBA) fills I^- and FA^+ vacancies through its acetate and guanidine groups, respectively, reducing H_2O binding and blocking perovskite hydration during ambient-air crystallization[148] (Fig. 4a). A complementary strategy relies on in situ formation of protective interphases during air processing. These approaches exploit intermediate-phase engineering or ion exchange reactions to generate self-encapsulating layers, such as PbI_2-rich shells[152] or molecular protective layers[153–155], that shield the perovskite from further moisture exposure while simultaneously passivating defects (Fig. 4b). This structural protection enables stable phase evolution and high device performance even at around 60% relative humidity[152].

Beyond strategies aimed at blocking moisture, ambient water can also be harnessed as a chemical reagent during perovskite crystallization. In these approaches, water-reactive modifiers react with ambient moisture to generate stabilizing species (such as maleic acid formed by hydrolysis of maleic anhydride[156]) in situ, which passivate defects, suppress phase transitions and form hydrophobic grain-boundary barriers[156,157]. By transforming moisture from a degradation factor into a functional component of crystallization, such strategies broaden the humidity tolerance window of air processing.

Strategies to increase the phase stability of the perovskite can also increase tolerance to ambient air, particularly moisture. Such strategies

include trace CdI_2 doping to relax lattice strain[158] and *n*-heptylamine-assisted intermolecular exchange to control the preferred crystallization orientation[159]. By stabilizing the desired perovskite phase and reducing defect-assisted degradation pathways, these approaches can facilitate scalable air fabrication with reduced reliance on additional protective layers during processing.

Solvent environment. Commonly used polar solvents such as DMF and DMSO are highly hygroscopic and volatile. Therefore, moisture uptake in ambient air can alter the coordination and evaporation kinetics of these solvents. These changes often cause inhomogeneous intermediate-phase formation, residual PbI_2 domains and pinhole defects during perovskite film formation, compromising device efficiency and stability.

The regulation of the precursor–solvent environment is an effective strategy to enable more controlled perovskite crystallization under air-processing conditions. For example, strongly coordinating molecules such as *N*,*N*-methylenebisformamide (NMF)[160] and pseudohalide anions[161] that bind Pb^{2+} ions or interact with organic cations are introduced into precursor inks. These strategies substantially enhance the stability of precursor complexes and intermediate phases, suppressing compositional fluctuations.

Another widely adopted approach involves modulating solvent volatility and water affinity. Replacing highly hygroscopic or rapidly evaporating solvents with alternatives that exhibit lower water solubility and slower evaporation rates, such as isobutanol (IBA), creates a more robust crystallization environment under ambient humidity. Such solvent selection strategies suppress unwanted phase transitions and the formation of residual PbI_2 domains across a broad humidity range, improving reproducibility and tolerance to environmental fluctuations, although the IBA dosage must be optimized to avoid increased residual PbI_2 (ref. 162).

Antisolvent treatment and precursor-additive passivation strategies have also been adapted to air processing. Examples include introducing functional additives such as two-dimensional MBene nanosheets through the antisolvent[163] or adding precursor-compatible nanoparticles to the PPSs[164]. These additives regulate nucleation, prolong crystal growth and provide simultaneous bulk and interfacial passivation, increasing the tolerance of perovskite film formation to ambient-air processing conditions.

Green solvents. To support ambient-air fabrication of PSCs, increasing efforts have focused on the development of solvent systems with reduced toxicity while maintaining crystallization control and process robustness under humid conditions. For example, biomass-derived γ-valerolactone (GVL) combined with coordinating additives such as 1,3-dimethyl-2-imidazolidinone (DMI) provides a workable solvation environment and enables nucleation and growth regulation[165]. Another option is low-toxicity alcohol-based systems such as *n*-butanol (nBA) that facilitate controlled film formation under ambient conditions owing to their low polarity, limited moisture uptake and moderate evaporation dynamics[166].

The feasibility of water-based processing has also been explored as the ultimate green solution. Using surfactants or surface-tension modifiers to regulate wetting and film formation enabled the formation of dense and pinhole-free perovskite layers from aqueous systems under dry-air conditions (relative humidity <10%)[167]. This example suggests that aqueous processing could be a feasible future low-toxicity alternative to hazardous organic solvents.

Reduced thermal budget

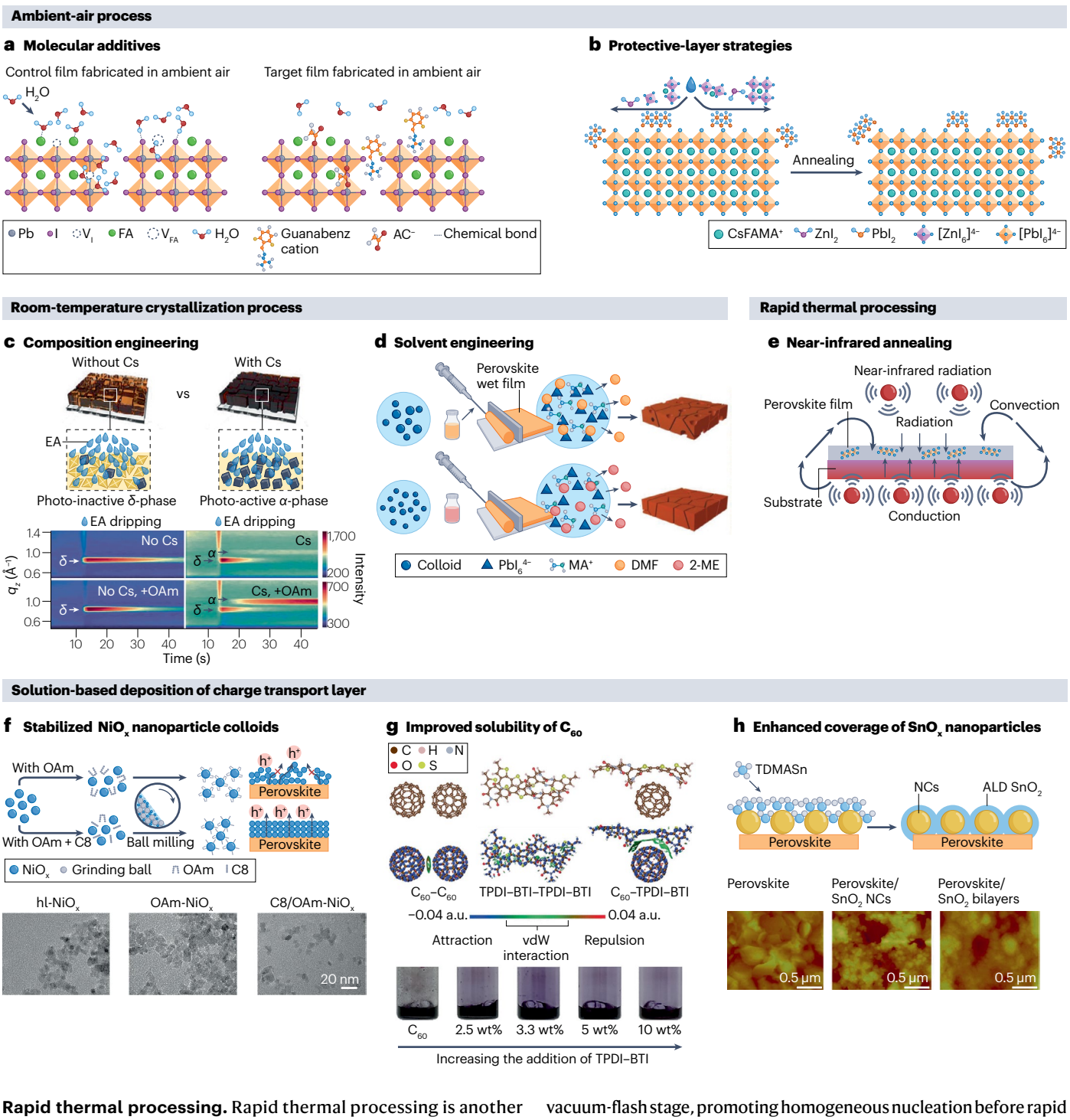
High-temperature annealing is widely regarded as essential for perovskite film fabrication, as it promotes phase-pure crystallization and good film morphology. However, prolonged heating at elevated temperatures imposes substantial energy demands, increases process complexity and contributes substantially to manufacturing cost. Accordingly, reducing the overall thermal budget by lowering processing temperature or shortening annealing duration can reduce energy consumption and manufacturing cost but at the expense of increased challenges in crystallization regulation, film formation and device stability.

Room-temperature process. Room-temperature perovskite film fabrication has emerged as an active research direction aimed at avoiding conventional high-temperature annealing during film formation.

A primary challenge in room-temperature processing arises from the thermodynamic instability of the photoactive black α-phase in FA-rich perovskites, which readily transforms into the non-photoactive δ-phase at low temperatures[168–170]. To overcome this limitation, composition engineering and facet-stabilization strategies have been widely adopted[168–170]. For example, partial substitution of FA^+ with smaller A-site cations such as Cs^+ stabilizes the Pb–I framework, expels residual solvent species and suppresses δ-phase formation, enabling direct room-temperature crystallization of the α-phase[168]. In parallel, molecular additives introduced into PPSs, such as ethylene thiourea (ETU)[169] and oleylamine[170], can selectively interact with specific crystal facets or undercoordinated surface sites to reduce surface free energy, suppress stacking-fault formation and lower the energetic barrier for crystallization (Fig. 4c). These effects reduce the energetic disadvantage of the α-phase relative to competing non-photoactive phases, facilitating its formation and stabilization at room temperature.

Beyond phase stability, insufficient crystallization kinetics is another major barrier for room-temperature processing because incomplete solvent removal often leaves residual intermediates and defect-rich films. Solvent and additive engineering can facilitate solvent removal and accelerate nucleation and phase conversion without external heating[171,172]. For example, replacing DMF with 2-ME produces smaller, more uniformly dispersed precursor colloids, promoting homogeneous nucleation and rapid crystallization at room temperature[171] (Fig. 4d). Additionally, strategies such as mixed-solvent-vapour annealing[173], the incorporation of volatile amines into the perovskite lattice to form transient low-dimensional intermediates[174] or the use of volatile, low-boiling-point solvent systems assisted by gas-quenching[175] can accelerate solvent removal, rapidly build supersaturation and promote spontaneous conversion into the 3D perovskite phase. These approaches enable fast crystallization within seconds to minutes at room temperature, yielding dense, highly crystalline films without subsequent high-temperature thermal annealing.

Film morphology constitutes a further challenge for room-temperature processing, as rapid nucleation and non-uniform solvent evaporation frequently induce pinholes, rough surfaces and high defect densities. To address this issue, PPS modification strategies have been developed to regulate ink–substrate interactions and crystallization. For example, multifunctional surfactants such as tetradecyldimethyl(3-sulfopropyl)ammonium hydroxide inner salt (TAH) are incorporated into PPSs to improve wetting on hydrophobic substrates, regulate nucleation dynamics and promote oriented grain growth, suppressing pinhole formation and reducing defect densities[176].



Rapid thermal processing. Rapid thermal processing is another way to reduce the energy consumption and processing time compared with conventional thermal annealing. One approach is to use near-infrared irradiation to achieve rapid heating through combined radiation, convection and conduction rather than relying primarily on heat conduction as in conventional hot-plate annealing[177] (Fig. 4e). To overcome incomplete nucleation and non-uniform growth caused by the ultrafast heating in this approach, 10% excess of PbI_2 was incorporated into the PPS to provide additional nucleation sites during the vacuum-flash stage, promoting homogeneous nucleation before rapid crystallization[177]. Laser annealing provides even more localized and intense energy delivery, enabling rapid crystallization before substantial ambient degradation occurs. For example, nanosecond laser irradiation supplies an irradiance two orders of magnitude higher than that of thermal annealing, achieving full perovskite phase conversion within approximately 20 s over a 100 cm^2 area[178]. Intense pulsed light annealing is another rapid processing strategy, in which millisecond-scale photothermal excitation volatilizes residual solvents and triggers fast

Fig. 4 | Strategies to simplify the fabrication of perovskite solar cells. **a**, The addition of guanabenz acetate salt (right) reduces moisture sensitivity by eliminating iodide vacancies (V_I) and formamidinium vacancies (V_{FA}), preventing perovskite decomposition triggered by the absorption of water molecules (left). **b**, Treatment with ZnI_2 in acetonitrile forms a PbI_2 shell on the perovskite surface through in situ cation exchange, passivating surface defects and protecting the perovskite from air exposure. Subsequent annealing accelerates solvent volatilization. **c**, Time-resolved grazing incidence wide-angle X-ray scattering measurements of perovskite films with and without Cs and oleylamine (OAm), showing that Cs enables the room-temperature transition from the photo-inactive δ-phase to the photoactive α-phase following the addition of ethyl acetate (EA) antisolvent, with OAm further accelerating this transition in the Cs-containing films. **d**, The use of 2-methoxyethanol (2-ME) as an alternative solvent to *N*,*N*-dimethylformamide (DMF). DMF predominantly interacts with Pb–I octahedral species and is associated with relatively large precursor colloids, whereas 2-ME interacts more strongly with MA^+ and produces much smaller and more uniformly dispersed colloids that favour homogeneous nucleation and room-temperature crystallization. **e**, Near-infrared annealing of perovskite films, combining radiation, conduction and convection for rapid and efficient heat transfer. **f**, Transmission electron microscopy images of hydrophilic NiO_x nanoparticles (hl-NiO_x), NiO_x with long-chain OAm ligands (OAm-NiO_x) and OAm-NiO_x with partial substitution of OAm with shorter-chain octylamine (C8/OAm-NiO_x). The dual-ligand approach suppresses nanoparticle aggregation and yields smaller particles, enabling the formation of a dense assembled hole transport layer (HTL) on the perovskite for efficient charge transport. **g**, Top, molecular structures and interaction region indicator (IRI) maps of C_{60}–C_{60}, TPDI–BTI–TPDI–BTI (dithienylpyrazinediimide (TPDI) and bithiophene imide (BTI)) and C_{60}–TPDI–BTI systems. Bottom, images of mixed TPDI–BTI: C_{60} solutions with increasing TPDI–BTI content, showing that TPDI–BTI–C_{60} interactions suppress C_{60} aggregation, enabling solution-processed C_{60} electron transport layers (ETLs). **h**, Atomic force microscopy images of a perovskite layer, a perovskite layer topped with SnO_2 nanocrystals (NCs) and the perovskite with the SnO_2 NCs topped with an atomic layer deposition (ALD) SnO_2 layer (using a TDMASn, tetrakis-(dimethylamino)tin(IV) precursor). The ALD SnO_2 overlayer fills defects and smooths the SnO_2 NC layer. AC^-, acetate; CsFAMA, Cs^+, formamidinium (FA^+) and methylammonium (MA^+) A-site cations; q_z, out-of-plane component of the scattering vector; vdW, van der Waals. Panel **a** reproduced from ref. 148, Springer Nature Limited. Panel **b** reproduced with permission from ref. 152, Wiley. Panel **c** reproduced with permission from ref. 170, Elsevier. Panel **d** reproduced with permission from ref. 171, Wiley. Panel **e** reproduced from ref. 177, CC BY 4.0. Panel **f** reproduced with permission from ref. 180, RSC. Panel **g** reproduced from ref. 181, CC BY 4.0. Panel **h** reproduced with permission from ref. 95, ACS.

nucleation and grain growth. In this approach, sacrificial coordination strategies based on removable Lewis-base additives were required to mitigate non-uniform crystallization and pinhole formation associated with such rapid heating[179].

Vacuum-to-solution deposition

In the manufacturing of PSCs, CTLs are commonly deposited by vacuum-based techniques to ensure conformal, uniform coverage and reliable electronic properties, especially for metal oxides and fullerenes, substantially increasing equipment requirements and energy consumption. Consequently, replacing vacuum-based deposition with solution processing offers a potential pathway to simplify fabrication, which could lower manufacturing cost by approximately 10–20% if comparable device performance can be maintained[14,93].

Solution-processed NiO_x in the form of nanoparticle dispersions is a promising substitute for HTMs prepared through magnetron sputtering. However, it is difficult to achieve a favourable Ni oxidation state (that is, a high Ni^{3+} fraction) and continuous film coverage from NiO_x nanoparticle inks because the Ni valence is sensitive to nanoparticle surface chemistry and particle aggregation compromises dispersion and film coverage. Hydrogen peroxide (H_2O_2) treatment can overcome this challenge by simultaneously regulating Ni oxidation states and promoting uniform, dense film formation from solution[47]. Additionally, surface-ligand regulation strategies have been explored to modulate interparticle interactions and ink stability, enabling a balance between colloidal stability in solution and efficient charge transport in the resulting films. For example, a dual-ligand strategy combining long-chain oleylamine for colloidal stabilization with shorter octylamine suppresses nanoparticle aggregation while reducing ligand-induced barriers to interparticle charge transport[180] (Fig. 4f).

For ETMs, solution-based deposition of C_{60} remains challenging owing to its poor solubility and tendency to aggregate. One approach to overcome this problem is to use functional polymers to solubilize and electronically couple C_{60} (Fig. 4g), forming uniform polymer–fullerene composite ETLs that can be deposited from solution[181]. The development of soluble C_{60} derivatives[182,183] or non-fullerene materials[184] provides an alternative materials-based route, enabling direct solution processing while maintaining comparable electronic properties to fullerene ETLs. Solution-processed inorganic ETLs, particularly those based on SnO_2, have been explored as alternatives to vacuum-deposited CTLs. However, as highlighted above, achieving uniform film formation and suitable electronic properties under the limited thermal budget compatible with perovskite layers remains challenging[94]. Consequently, hybrid strategies that incorporate ultrathin ALD-grown interfacial layers alongside solution-processed SnO_2 nanocrystal layers have been used to improve film conformity and interfacial characteristics[95] (Fig. 4h). Further work is thus needed to develop fully solution-processable SnO_2 ETLs to better realize their cost advantages.

Conclusion and outlook

This Review highlights three complementary cost-driven pathways towards industrially manufacturable perovskite PVs: increasing manufacturing yield, reducing material cost and simplifying fabrication. Although substantial progress has been achieved along each pathway, distinct challenges remain, as detailed below.

Manufacturing yield remains a bottleneck for the commercialization of PSCs, as small variations in material purity, PPS stability and fabrication processes can propagate and amplify along the manufacturing chain, leading to poor reproducibility and pronounced material waste. Suppressing such variability at its source is essential for yield improvement. Future research should place greater emphasis on systematic investigations into how defects are introduced, propagated and accumulated across the manufacturing chain. Additionally, standardized tolerance evaluation criteria with clearly defined parameter tolerance ranges should be established to increase reproducibility across different manufacturing environments. Ultimately, prioritizing material formulations and fabrication processes with high tolerance to variability rather than relying solely on increasingly stringent material purity and process control requirements will be essential for achieving reproducible, scalable and low-cost PSC manufacturing.

Material cost reduction can be pursued through three complementary strategies: material substitution, structural simplification and recycling. Although advances have been made in these areas, low-cost materials and simplified device architectures still generally have lower

efficiency and stability than their state-of-the-art counterparts. Future research should therefore focus on developing low-cost materials with competitive performance through deposition-process simplification, multifunctional material design and interfacial engineering, paying particular attention to cost-dominant functional layers such as TCOs and C_{60}. Additionally, further progress in developing CTL-free devices will depend on restoring charge selectivity and interfacial stability at perovskite–electrode interfaces through targeted interfacial modification. Scalable recycling routes are still needed because existing studies are largely limited to laboratory-scale demonstrations, and it remains unclear whether their high recovery ratios can be maintained under mass-production conditions. Unresolved challenges in recovered-material purity, solvent recovery and reuse, and cross-layer compositional diffusion further mean that the economic viability of industrial-scale PSC recycling has yet to be established.

Fabrication simplification offers a direct pathway to reduce energy consumption and equipment requirements if comparable device performance and operational stability can be maintained. However, simplified fabrication faces challenges in terms of scalability and manufacturing robustness. High tolerance to environmental fluctuations is needed to enable ambient-air processing. Additionally, low-energy thermal treatments result in narrower processing windows for crystallization, requiring tighter control over nucleation and solvent-removal kinetics. Solution-processed CTLs remain limited by challenges in material dispersibility, film uniformity and electronic properties under reduced thermal budgets, which make it difficult to maintain device performance and operational stability comparable to those achieved with vacuum-deposited CTLs. Overall, fabrication simplification should not be viewed as a mere removal or substitution of existing steps but rather as a holistic redesign aimed at reducing energy and equipment demands while preserving device performance.

Research on cost-reduction strategies for PSCs still largely focuses on individual cost-related dimensions and the effects of these strategies on device performance, operational stability and manufacturing compatibility. Future studies should aim to develop integrated cost-reduction strategies that simultaneously address manufacturing yield, material cost and fabrication complexity. Achieving a balance among these dimensions under realistic industrial conditions will be important for advancing low-cost PSC technologies towards large-scale commercialization.

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Author contributions

Z.Z., Y.L., Z.Q. and L.H. researched data for the article. Z.Z., M.C., Y.W. and L.H. contributed substantially to discussion of the content. Z.Z. wrote the article. All authors reviewed and/or edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

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